710001 POWER UNIT - in panel B2 (factory wired and tested; earthed aluminium enclosure)

INLET (outside B2)
powerCON TRUE1 TOP
16 A 250 V, locking
v1 position, X 120 Z 313

L N PE

RCBO 1P+N
C10, 30 mA
the only mains
protection

USER SOCKET
UTEN S300 (plan VT07)
on the unit face,
Z 1000 (PM ruling)

W01 3x1.5

W02 L N

PE TERMINAL
inlet PE, socket PE,
enclosure stud,
B2 frame bond W03

PS1 60 W
12 V 5 A, Class 2
input 100-240 V
Class II (no PE)

F1 5 A

W03 2.5 mm2 G/Y to B2's frame
(only B2 carries a mains part)

W02 L N

LAN
etherCON NE8FDP in B2,
W04 Cat6 to port 1
port 2: W31 to the kit

PS2 60 W DIMMABLE
12 V PWM, Class 2
input 100-240 V
Class II (no PE)

PS2 out

F2 3 A

DIM 0-10 V

W02 L N

PS3 60 W
12 V 5 A, Class 2
input 100-240 V
Class II (no PE)

port 2

JOINTS AND HARNESS SEGMENTS (plug pair at every joint; cables clip to the cassettes only, D-26)

X1 unit / B2-CASS | X2 corner, in R1-CASS | X3 into the air column

W30b 4x18 AWG B2-CASS

W30c R1-CASS

X7 VOICE KIT DROP (R1, behind the screen)
12 V 5 A outlet: Mini-Fit Jr 4-way, capped
Cat6 keystone, capped; draw wire W32
reached through felt plug FA-KIT

W10b B2-CASS

X4

W10c B4-CASS

X5

W11 in the shelf

LIGHT SHELF (back and side wall tops, D-70)
Loox5 2168 strip 4.43 m, 26.6 W, uplight onto the felt
fed at X 355 on the back wall: back strip both ways,
jumpers to the left and right wall strips

W20b air feed 4x18 AWG (2 per pole)

W20c R1-CASS

W20d column

W20 and W24 share one 10-way plug at X1, X2 and X3

W24b hub link 6x24 AWG: DIM, light current, rails

W24c

W24d

W31b Cat6 (RJ45 couplers)

W31c to the kit drop X7

AIR COLUMN (front-right inside corner; hub zone 180 x 180 at Z 1200, D-61, D-70)

FUSE BLOCK
behind the hub zone cover
F3a 2 A supply fans
F3b 2 A exhaust fans
F3c 1 A hub
F3d 1 A call light

DESK PRO AI SENSOR HUB
in-house, cited by name; CO2, presence
out: PWM 1, 2; DIM 0-10 V; CALL
in: tach 1, 2; inlet sensor; buttons;
light currents; PS1-PS3 rails (H-ELE-05)

SUPPLY FANS S1 S2
high in DR; W40 4 x 20 AWG

EXHAUST FANS E1 E2
soffit right end; W41

CONTROLS (hub zone)
BOOST, TEST, CALL buttons
LIGHT knob: all hub inputs
(W44); TEST = self-test D-87

CALL LIGHT (R2, above the screen)
900 mm bar, 2.7 W; hub switches and
senses it; W12a column, W12b R2-CASS
X6 behind the column

INLET TEMP SENSOR
supply fan box inlet; W42

SAFETY AND RESERVES
SELF-TEST (D-87): TEST checks the 3 rails, runs both fan pairs (tach), cycles both lights (current), reads the sensors and shows PASS or the failing item. PWM fail-safe: an open PWM line runs the fans at full speed. Draw wires: W32 kit drop -> column -> soffit (mic, status light); W33 back wall top -> ceiling cassette (D-41, and the sprinkler or alarm wiring point, ISS-052). Nothing is cut on site later (REQ-VCK-03).

NOTES

1. Mains (red) exists only inside B2 and the power unit (D-10, IF-12). Nothing mains is connected on site.
2. Every 12 V circuit (blue) is Class 2: each supply is 60 W, the Class 2 limit at 12 V (compliance note).
3. Plug pairs: supply side receptacle (female), load side plug (male); Molex Mini-Fit Jr, keyed by way count.
4. Every cable through sheet steel runs in a grommet; cables clip to the cassette tray only (D-26).
5. Wire sizes and lengths: harness list (sheet 2 and the _SPEC.csv). Voltage drop under 3 %: calc-results.md.
6. Bought parts are on HOLD: no order, no supplier contact. The in-house hub is cited by name only.
7. US version: change the cord set, the inlet side and the socket only; supplies take 100-240 V.
8. Joints: X1 unit/B2-CASS X2 corner (R1-CASS) X3 column entry X4 B2/B4 X5 shelf X6 call X7 kit.

PROJECT	FocusPod v2 - Spaceship
TITLE	**ELECTRICAL - WIRING DIAGRAM**
DWG / PART NO.	710000 ELECTRICAL
REV / STATUS	P01 / DRAFT - not for production
UNITS / SCALE	mm / not to scale (schematic)
SHEET	1 of 3
DATE / BY	2026-10-31 / ELECTRICAL Engineer
SOURCE	electrical_model.py, electrical_parts.py

HARNESS LIST (every field, with conductors, routes, bushings and voltage drop: 710000_P01_ELECTRICAL-HARNESS-LIST_SPEC.csv)

ID	SEGMENT	FUNCTION	CABLE	L mm	END A	END B
W01	PU	mains feed	H07V-K 1.5 mm2 single cores in the sleeve	920	inlet terminals (powerCON TRUE1 TOP)	RCBO, PE terminal
W02	PU	mains inside the unit	H05V-K 1.0 mm2	1500	RCBO	socket, supply input leads
W03	PU	protective bond	H07V-K 2.5 mm2 green-yellow	300	M6 ring lug	M6 ring lug
W04	PU	LAN in	Cat6 U/UTP patch cord	870	RJ45 plug into the feed-through	punched down at the socket
W10a	PU	light feed	2 x 18 AWG, Class 2 (CL2 or equivalent)	300	PS2 output terminals	Mini-Fit Jr 2-way receptacle
W10b	B2-CASS	light feed	2 x 18 AWG, Class 2	270	Mini-Fit Jr 2-way plug (mates W10a)	Mini-Fit Jr 2-way receptacle (tail out of the top grommet)
W10c	B4-CASS	light feed	2 x 18 AWG, Class 2	1360	Mini-Fit Jr 2-way plug (mates W10b in P4)	Mini-Fit Jr 2-way receptacle, tail into the shelf
W11a	SHELF	light shelf feed lead	2 x 18 AWG, Class 2	300	Mini-Fit Jr 2-way plug (mates W10c)	strip feed: back strip both ways
W11b	SHELF	jumper to the left wall strip	2 x 18 AWG, Class 2	1620	strip feed connector	strip feed connector
W11c	SHELF	jumper to the right wall strip	2 x 18 AWG, Class 2	910	strip feed connector	strip feed connector
W20a	PU	air feed	4 x 18 AWG (2 per pole), Class 2	300	PS3 output	Mini-Fit Jr 10-way receptacle (with W24)
W20b	B2-CASS	air feed	4 x 18 AWG (2 per pole), Class 2	910	Mini-Fit Jr 10-way plug (with W24)	Mini-Fit Jr 10-way receptacle (with W24), tail through the corner slot S-CNR
W20c	R1-CASS	air feed	4 x 18 AWG (2 per pole), Class 2	1260	Mini-Fit Jr 10-way plug (with W24) (mated in P2 through the felt slot FS-CNR)	Mini-Fit Jr 10-way receptacle (with W24), tail into the column
W30a	PU	voice kit feed	4 x 18 AWG (2 per pole), Class 2	300	F1 5 A	Mini-Fit Jr 4-way receptacle
W30b	B2-CASS	voice kit feed	4 x 18 AWG (2 per pole), Class 2	910	Mini-Fit Jr 4-way plug	Mini-Fit Jr 4-way receptacle, tail through the corner slot S-CNR
W30c	R1-CASS	voice kit feed	4 x 18 AWG (2 per pole), Class 2	1050	Mini-Fit Jr 4-way plug (mated in P2 through the felt slot FS-CNR)	Mini-Fit Jr 4-way receptacle, capped with an empty plug housing (the kit outlet, 12 V 5 A)
W24a	PU	hub link to the power unit	6 x 24 AWG screened, Class 2	300	PS2 DIM input, shelf current sense, PS1 and PS2 rail taps	shares W20's 10-way receptacle
W24b	B2-CASS	hub link to the power unit	6 x 24 AWG screened, Class 2	910	shares W20's 10-way plug	shares W20's 10-way receptacle, tail through the corner slot S-CNR
W24c	R1-CASS	hub link to the power unit	6 x 24 AWG screened, Class 2	1260	shares W20's 10-way plug (mated in P2 through the felt slot FS-CNR)	shares W20's 10-way receptacle
W31a	PU	voice kit data	Cat6 U/UTP	300	LAN socket port 2	RJ45 inline coupler
W31b	B2-CASS	voice kit data	Cat6 U/UTP	910	RJ45 inline coupler	RJ45 inline coupler, tail through the corner slot S-CNR
W31c	R1-CASS	voice kit data	Cat6 U/UTP	1050	RJ45 inline coupler (in P2)	RJ45 keystone jack, capped (the kit's data link)
W20d	COLUMN	air feed	4 x 18 AWG (2 per pole), Class 2	420	Mini-Fit Jr 10-way plug (with W24d), mates W20c behind the column	fuse block input
W24d	COLUMN	hub link to the power unit	6 x 24 AWG screened	420	shares W20d's 10-way plug	hub I/O (HOLD H-ELE-05)
W40	COLUMN	supply fan pair	4 x 20 AWG	1580	fuse block and hub terminals	4-way PWM fan connector into a Y-cable at the fan pair (housing HOLD H-ELE-07)
W41	COLUMN	exhaust fan pair	4 x 20 AWG	1260	fuse block and hub terminals	4-way PWM fan connector into a Y-cable (housing HOLD H-ELE-07)
W42	COLUMN	inlet temperature sensor	4 x 24 AWG shielded	1780	hub sensor connector (HOLD, ISS-060)	sensor probe (HOLD H-ELE-06)
W43	COLUMN	hub supply	2 x 22 AWG	300	fuse block	hub power connector (HOLD, ISS-060)
W44	COLUMN	controls loom	6 x 24 AWG screened	300	button and knob tags	hub inputs (HOLD H-ELE-05)
W12a	COLUMN	call light	2 x 20 AWG, Class 2	550	hub switched output (or the I/O board's switch, H-ELE-05)	Mini-Fit Jr 2-way receptacle, tail out of the column
W12b	R2-CASS	call light	2 x 20 AWG, Class 2	570	Mini-Fit Jr 2-way plug (mates W12a)	strip feed of the call bar
W32	R1-CASS + COLUMN	draw wire: kit zone to the soffit (mic, st..	nylon draw cord 3 mm	1870	tied at P3-KIT	tied at the soffit entry
W33	B4-CASS + CEILING	draw wire: ceiling route (D-41)	nylon draw cord 3 mm	1680	tied at P6	tied at the ceiling cassette's front hole

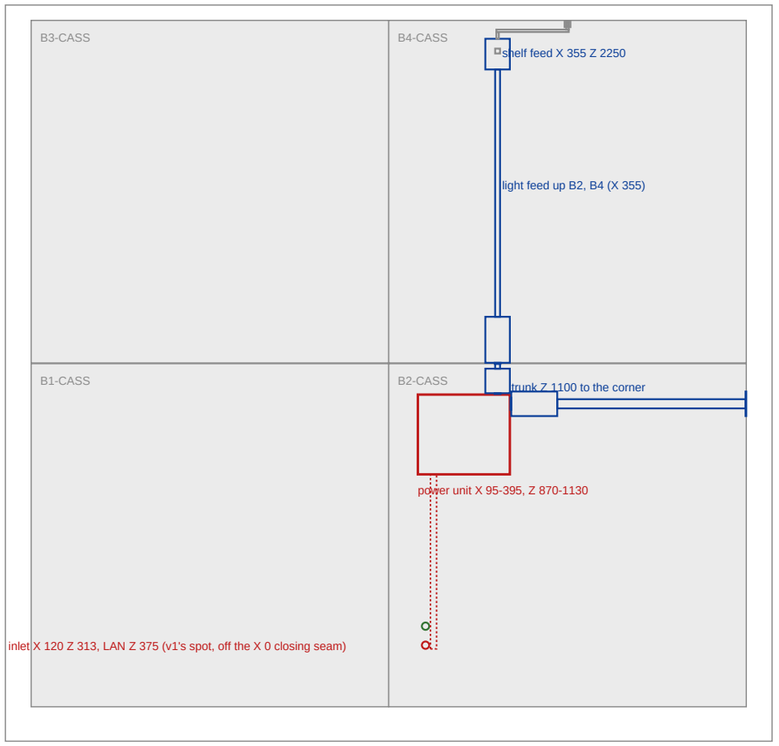
NOTES

1. Lengths: route model (calculation) + 150 mm loops and tails (assumption). The kit drop has a 500 mm service loop.
2. Plug pairs: supply side receptacle (female), load side plug (male). W20 and W24 share one 10-way at X1, X2, X3.
3. Every segment is tested in the factory before it ships (D-87): continuity, polarity, no shorts, latch pull.
4. Cables: Class 2 circuits on 300 V PVC cable (CL2 or CL3 listed for the US phase); mains only inside B2 and the unit.
5. W32 and W33 are draw cords, tied off at each hole, for the voice kit's later cables (REQ-VCK-03).

PROJECT	FocusPod v2 - Spaceship
TITLE	**ELECTRICAL - HARNESS LIST**
DWG / PART NO.	710000 ELECTRICAL
REV / STATUS	P01 / DRAFT - not for production
UNITS / SCALE	mm / table, not to scale
SHEET	2 of 3
DATE / BY	2026-10-31 / ELECTRICAL Engineer
SOURCE	electrical_model.py, electrical_parts.py

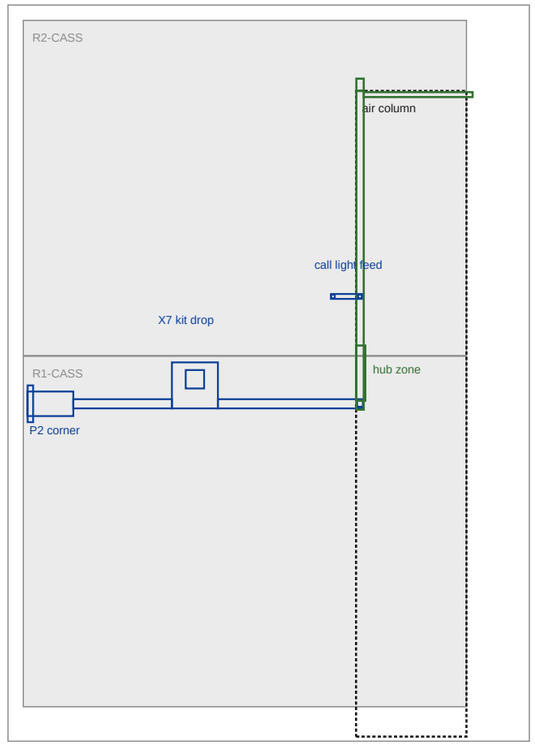
VIEW A - BACK WALL FROM INSIDE, looking toward +Y (1:20)

X view = +X, Y view = +Z; origin O on the room floor at the pod centre



VIEW B - RIGHT WALL FROM INSIDE, looking toward +X (1:20)

X view = -Y (front to the right), Y view = +Z



RESERVED SPACES IN THE CAD

Routes, pockets, bushings and slots are reserved spaces in the cassettes' 24 mm gap, in 710000_P01_ELECTRICAL_CAD, derived from skeleton P04. Pockets: P1 at the unit, P2 in R1-CASS behind the felt slot FS-CNR (hidden by B2-CASS's end), P4 above the B2/B4 joint, P5 and P6 the light feed, P3 the voice kit drop.

SITE ORDER (two people, one 5 mm key, no electrician)

1 Hang R1-CASS and R2-CASS; their tails stay out.
2 Hold B2-CASS near: plug its leads to the unit (X1) and to R1's corner tails (X2), push the pairs into P1 and P2, hang it.
3 B4-CASS: plug X4, push the pair into P4, hang it.
4 Plug the light shelf at X5 when the shelf goes up.
5 Before the air column goes in, plug X3 and X6 behind it.
6 Plug the fans at their boxes (DR inner face; soffit underside).
7 Room cord into the inlet, RCBO on, press TEST: PASS (D-87).

KEEP-OUTS FOR OTHER TEAMS

WALL: B2 bay X 95-395, Z 870-1130 for the unit; sleeve X 136-156, Z 301-870; the holes and slots in B2-CASS, B4-CASS, R1-CASS, R2-CASS.
INTERIOR: ledge and screen fixings 15 mm off the trunk band Z 1085-1115.
CEILING: column chase 24 x 24 (X 1140-1164, Y -310 to -286), hub zone, soffit entry; the ceiling cassette's holes at X 583 (D-41 route).
FRONT: an opening in DR's back plate, Y -665 to -645, Z 2100-2116.

NOTES

1. Every coordinate in these views is computed from electrical_model.py (skeleton P04 parameters), so they follow the CAD.
2. Bushings: Heyco 2836 in a 19.1 mm hole (G-PU-TOP, G-B2-B4, FX-CEIL, G-CEIL-B/F); slots S-PU-X and S-CNR get an edge strip.
3. Felt-only holes (FX-*, FS-CNR, FA-KIT) need no bushing: no steel edge.
4. The Fusion drawing of the route model replaces these views when the drawing start is allowed (screen unlocked).

PROJECT	FocusPod v2 - Spaceship
TITLE	**ELECTRICAL - ROUTES AND JOINTS**
DWG / PART NO.	710000 ELECTRICAL
REV / STATUS	P01 / DRAFT - not for production
UNITS / SCALE	mm / 1:20 (views)
SHEET	3 of 3
DATE / BY	2026-10-31 / ELECTRICAL Engineer
SOURCE	electrical_model.py, electrical_parts.py